Letter of Transmittal

Date: 03-September-2025

To
Rubayea Ferdows
Assistant Professor
Computer Science and Engineering
IUBAT-International University of Business Agriculture and Technology

Subject: Submission of Project Report on *AgriLink – Online Shopping Cart*

Dear Madam,

It is my great pleasure to submit this project report titled *"AgriLink – Online Shopping Cart"* which has been prepared as part of the requirements of the [Course Name] course. This report represents my effort to design, analyze, and implement a database-driven e-commerce system where farmers can directly connect with urban buyers, thereby eliminating middlemen.

I sincerely hope that this report will fulfill your expectations and provide a clear understanding of the work I have done. I would like to thank you for your continuous guidance and support throughout the completion of this project.

Sincerely,
Md. Naimul Haque Sarker Galib (22303316),

Khaza Mezbah Uddin (22303321)

IUBAT-International University of Business Agriculture and Technology

Abstract

This project report presents the design and development of *AgriLink – an Online Shopping Cart for Agricultural Products*. The system connects **farmers directly with buyers**, ensuring fair prices for producers and affordable costs for consumers.

The project was developed using **PHP, MySQL, HTML, CSS, and XAMPP** as the development environment. The system incorporates three types of users: Farmers, Buyers, and Admin. Features include user authentication, product listing, cart and checkout, order management, feedback, and admin control.

The report provides an overview of system analysis, design, database modeling, implementation, and testing. It highlights the limitations of the existing manual system, the advantages of the proposed solution, and the feasibility of deploying such a platform in real-world scenarios.

Acknowledgement

First and foremost, I express my gratitude to the Almighty for giving me the strength and patience to successfully complete this project.

I would like to express my sincere appreciation to my course instructor Rubayea Ferdows, for their valuable guidance, constructive feedback, and encouragement throughout the project work.

I also extend my gratitude to my classmates and friends who provided useful suggestions, and to my family members for their continuous support and inspiration.

Finally, I acknowledge the contribution of online resources, documentation, and open-source tools that made this project possible.

Declaration

I hereby declare that the project report entitled *“AgriLink – Online Shopping Cart”* has been prepared by me as part of the requirements for the CSC-387 course. The work presented here is my original work, carried out under the supervision of **Rubayea Ferdows.**

I further declare that this project has not been submitted elsewhere, either partially or fully, for the award of any degree, diploma, or certificate. Any materials or concepts taken from other sources have been properly acknowledged and referenced.

Md. Naimul Haque Sarker Galib (22303316),

Khaza Mezbah Uddin (22303321)

03-September-2025

Letter of Authorization

This is to certify that the project report titled *“AgriLink – Online Shopping Cart”* submitted by Md. Naimul Haque Sarker Galib (22303316),Khaza Mezbah Uddin (22303321) has been carried out under my supervision and guidance as a partial requirement of the CSC-387 course at IUBAT-International University of Business Agriculture and Technology.

I hereby authorize the submission of this report for evaluation.

Supervisor’s Signature

Rubayea Ferdows
Assistant Professor
Computer Science and Engineering
IUBAT-International University of Business Agriculture and Technology

Assignment Report: AgriLink – Online Shopping Cart for Farmers & Buyers

Content Page

1. Introduction

1.1 Introduction of Project

This project, titled **AgriLink – Online Shopping Cart**, is a web-based platform where farmers can sell their agricultural products directly to urban buyers. The system eliminates middlemen, ensuring fair prices for farmers and affordable costs for buyers. The application is designed using **PHP, MySQL, HTML, and CSS**, and runs on **XAMPP** as the local server environment.

1.2 Aim of Project

- To build an online marketplace for agricultural products.
- To connect farmers directly with buyers.
- To provide admin controls for product, user, and order management.
- To implement user-friendly authentication and transaction workflows.

1.3 System Study & Analysis

1.3.1 System Analysis

The project follows structured system analysis:

- **Users:** Farmers, Buyers, Admin.
- **Functionalities:** Login, Signup, Product listing, Cart, Checkout, Payment, Feedback, Admin management.
- **Data Flow:** Farmers upload products → Buyers browse and purchase → Admin manages system.

2. Existing System

2.1 Existing Manual System

Currently, farmers sell through traditional markets, relying heavily on intermediaries.

2.2 Process of Existing System

- Farmers transport goods to markets.
- Middlemen purchase and resell products.
- Buyers purchase at inflated prices.

2.3 Problems with Existing System

- Farmers earn less profit.
- Buyers face higher costs.
- Limited transparency in transactions.

3. Proposed System

3.1 Aim of Proposed System

To build a **direct digital platform** between farmers and buyers.

3.1.1 Advantages

- Elimination of middlemen.
- Fair trade pricing.
- Faster transactions.
- Transparent records for admin.

3.2 System Feasibility Study

- **Technical:** Achievable using PHP, MySQL, and XAMPP.
- **Economic:** Low development cost, scalable for future.
- **Operational:** Easy to use, minimal training required.

4. Proposed System Design

4.1 Introduction of Proposed System

A three-user system:

- **Farmer Dashboard** (Add/update products).
- **Buyer Dashboard** (Search, filter, purchase).
- **Admin Dashboard** (Manage all users, products, orders).

4.1.1 Logical Design

- Entity Relationships: Users ↔ Products ↔ Orders.
- Logical rules for transactions.

4.1.2 Physical Design

- MySQL database with tables: Users, Products, Orders, Order_Items, Feedback.

4.1.3 Design/Specification Activities

- Web-based interface.
- Role-based access.
- Secure session handling.

5. Implementation of Model

5.1 Analysis Modeling & Design Methodologies

5.1.1 Database Design

Database: `agrilink`

Tables: `users`, `products`, `orders`, `order_items`, `feedback`.

5.1.2 Entity Relationship Model

Identifies entities: User, Product, Order, Feedback.

5.1.3 Identifying Entities

- **User (Farmer/Buyer/Admin)**
- **Product**
- **Order**
- **Feedback**

5.1.4 Entity Relationship Diagram (Introduction)

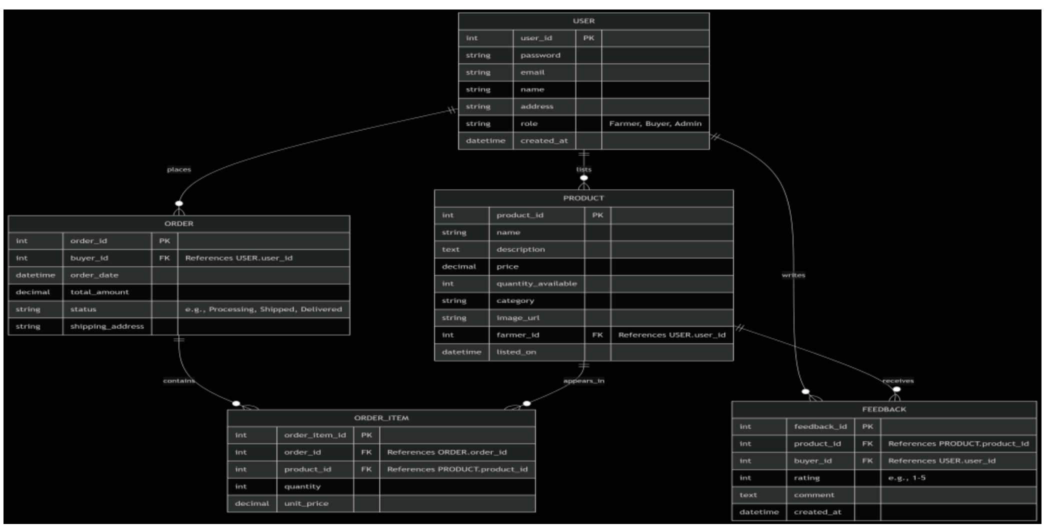
ER diagram shows links between users, products, and orders.

5.1.5 Relationship Cardinality

- A farmer → many products.
- A buyer → many orders.
- An order → many order_items.

5.1.6 Entity Relationship Diagram (Diagram)

- **Entity:** Rectangle
- **Attribute:** Listed inside the entity rectangle.
- **Relationship:** Diamond (optional in modern notation, implied by line).
- **Cardinality:** |
  - | = One
  - }o = Zero or more (Many)
  - || = Exactly one



5.1.7 Database Table Structure

DROP DATABASE IF EXISTS agrilink; CREATE DATABASE agrilink CHARACTER SET utf8mb4 COLLATE utf8mb4_general_ci; USE agrilink;

-- USERS CREATE TABLE users (id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100) NOT NULL, email VARCHAR(120) NOT NULL UNIQUE, password VARCHAR(255) NOT NULL, role ENUM('buyer','farmer','admin') NOT NULL DEFAULT 'buyer', phone VARCHAR(30), address VARCHAR(255), district VARCHAR(120), farm_name VARCHAR(150), is_verified TINYINT(1) NOT NULL DEFAULT 0, -- admin verifies farmers created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP);

-- PRODUCTS (owned by farmer) CREATE TABLE products (id INT AUTO_INCREMENT PRIMARY KEY, farmer_id INT NOT NULL, name VARCHAR(150) NOT NULL, category VARCHAR(100), description TEXT, unit_type ENUM('kg','liter','dozen','crate','bundle','piece') DEFAULT 'kg', price_per_unit DECIMAL(10,2) NOT NULL, stock_qty DECIMAL(10,2) NOT NULL DEFAULT 0, location VARCHAR(120), harvest_date DATE NULL, image_path VARCHAR(255), created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, FOREIGN KEY (farmer_id) REFERENCES users(id) ON DELETE CASCADE, INDEX (farmer_id), INDEX (name), INDEX (category));

-- CART (per buyer) CREATE TABLE cart (id INT AUTO_INCREMENT PRIMARY KEY, user_id INT NOT NULL, -- buyer id product_id INT NOT NULL, quantity DECIMAL(10,2) NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, UNIQUE KEY uq_cart (user_id, product_id), FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE, FOREIGN KEY (product_id) REFERENCES products(id) ON DELETE CASCADE);

-- ORDERS (placed by buyer) CREATE TABLE orders (id INT AUTO_INCREMENT PRIMARY KEY, buyer_id INT NOT NULL, total_amount DECIMAL(10,2) NOT NULL, payment_method VARCHAR(50) DEFAULT 'COD', payment_status ENUM('Pending','Paid','Failed') DEFAULT 'Pending', status ENUM('Pending','Confirmed','Cancelled','Shipped','Delivered') DEFAULT 'Pending', shipping_address VARCHAR(255), created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, FOREIGN KEY (buyer_id) REFERENCES users(id) ON DELETE CASCADE, INDEX (buyer_id), INDEX (status));

-- ORDER ITEMS (split revenue by farmer via product.farmer_id) CREATE TABLE order_items (id INT AUTO_INCREMENT PRIMARY KEY, order_id INT NOT NULL, product_id INT NOT NULL, farmer_id INT NOT NULL, -- denormalized for fast farmer reports unit_price DECIMAL(10,2) NOT NULL, quantity DECIMAL(10,2) NOT NULL, line_total DECIMAL(10,2) GENERATED ALWAYS AS (unit_price * quantity) STORED, FOREIGN KEY (order_id) REFERENCES orders(id) ON DELETE CASCADE, FOREIGN KEY (product_id) REFERENCES products(id) ON DELETE CASCADE, FOREIGN KEY (farmer_id) REFERENCES users(id) ON DELETE CASCADE, INDEX (farmer_id), INDEX (order_id));

-- FEEDBACK CREATE TABLE feedback (id INT AUTO_INCREMENT PRIMARY KEY, user_id INT NOT NULL, product_id INT NULL, rating TINYINT NULL, message TEXT NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, FOREIGN KEY (user_id)

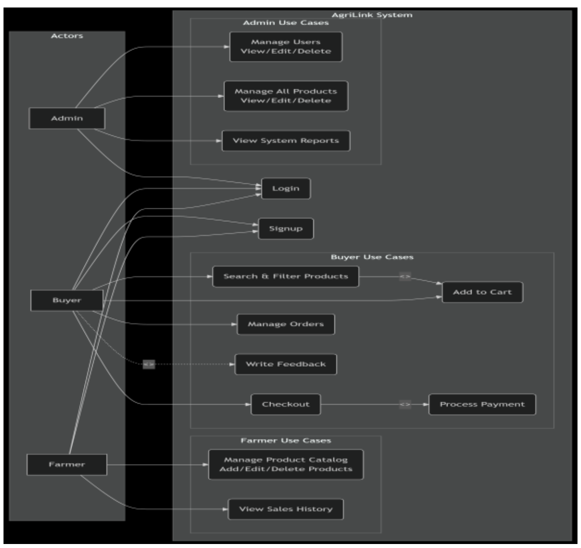
REFERENCES users(id) ON DELETE CASCADE, FOREIGN KEY (product_id) REFERENCES products(id) ON DELETE SET NULL);

-- ADMIN seed INSERT INTO users (name,email,password,role,is_verified) VALUES ('Site Admin','admin@agrilink.local', PASSWORD('admin123'), 'admin', 1);

-- DEMO farmers/buyers (optional) INSERT INTO users (name,email,password,role,phone,district,farm_name,is_verified) VALUES ('Green Field Farm','farmer1@agrilink.local', PASSWORD('farmer123'),'farmer','017xxxxxxx','Gazipur','Green Field',1), ('City Buyer','buyer1@agrilink.local', PASSWORD('buyer123'),'buyer','018xxxxxxx','Dhaka',NULL,1);

5.2 System Description

5.2.1 Data Flow Diagram



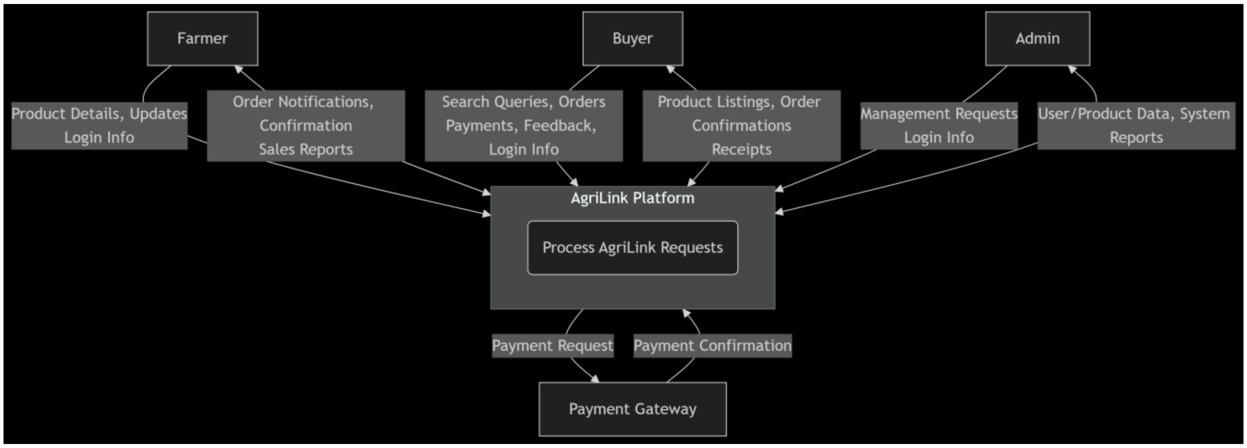
5.2.2 Symbol of DFD

- Circle → Process
- Arrow → Data flow
- Rectangle → External Entity
- Open Rectangle → Data store

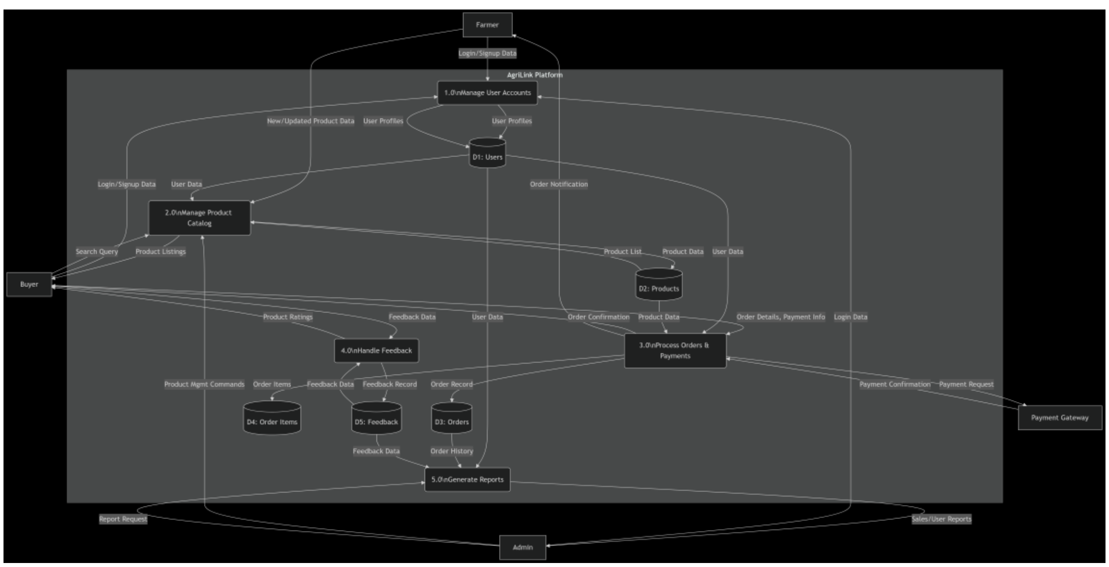
5.2.3 DFD of Project

Level 0: User login/signup → Product management → Checkout → Order confirmation.

Level 0 :



Level 1:



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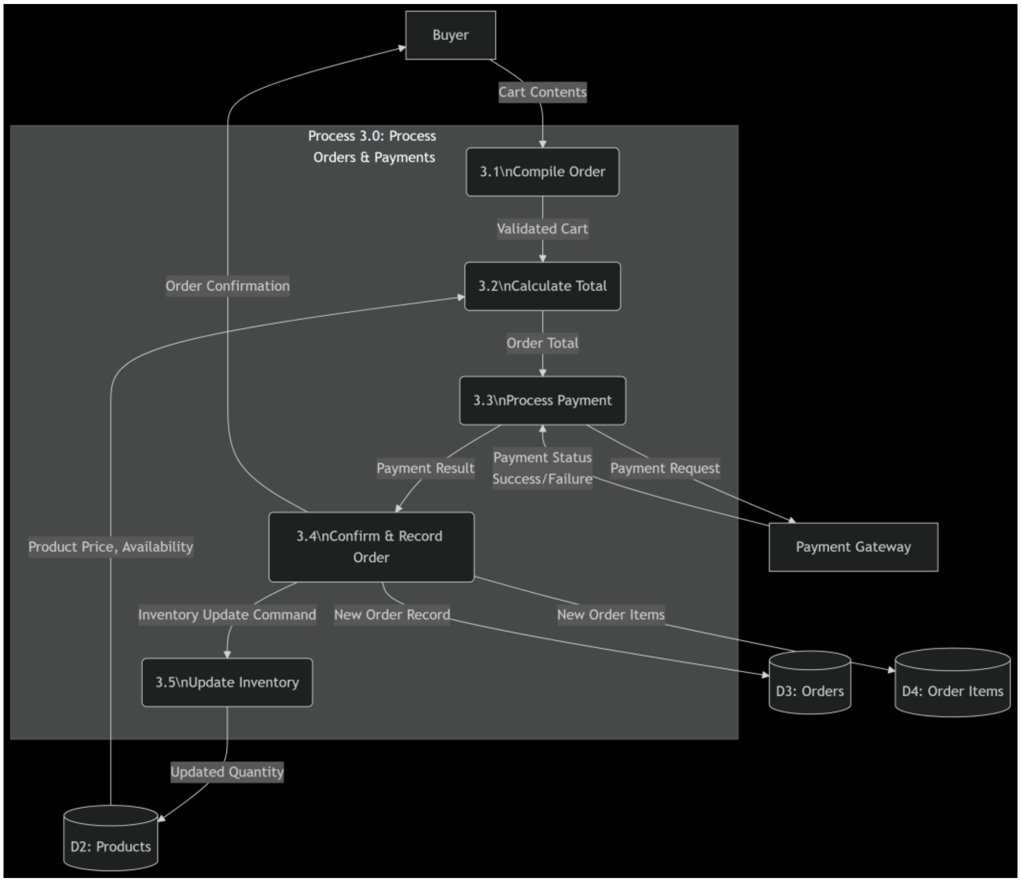


5.2.4 Effort Distribution

- Backend (PHP/MySQL): 40%
- Frontend (HTML/CSS): 30%
- Testing: 20%
- Documentation: 10%

5.2.5 Task Distribution

- Database setup
- User authentication
- Product listing & search
- Cart & Checkout
- Admin dashboard

5.2.6 Time Chart for Activities

- Week 1: Requirement analysis
- Week 2: Database setup
- Week 3–4: Backend coding
- Week 5: Frontend design
- Week 6: Testing and deployment

5.2.7 System Specification

Supports role-based access and CRUD operations.

5.2.8 Project Cost Estimation

5.2.8.1 Hardware Costs

Item	Quantity	Unit Price (BDT)	Total (BDT)
Developer Laptop/Desktop (Core i5, 8GB RAM, 500GB SSD)	1	60,000	60,000
Backup External HDD (1TB)	1	6,000	6,000
Networking (Router, Cabling, UPS)	1 set	8,000	8,000
Subtotal			**74,000**

5.2.8.2 Software Costs

(All software used is open-source / free, but approximate commercial equivalents are shown for real-world value.)

Item	License Type	Approx. Cost (BDT)
XAMPP (Apache, PHP, MySQL)	Free (Open Source)	0
Visual Studio Code / Sublime	Free	0
phpMyAdmin	Free	0
Windows 11 / Ubuntu OS	Pre-installed / Free	0
Commercial Alternatives (e.g., SQL Server, MS Visual Studio)	Proprietary	~25,000 (not used, but comparable value)

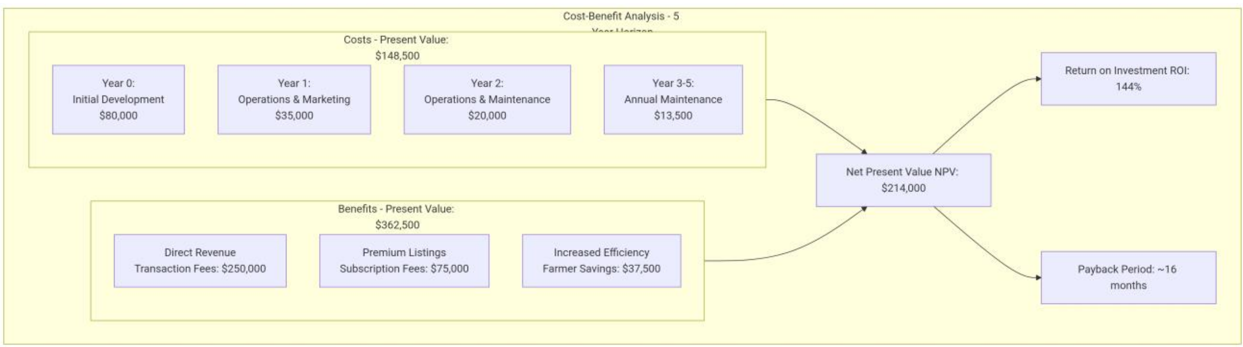
Subtotal: 0 BDT (since open-source stack was used).

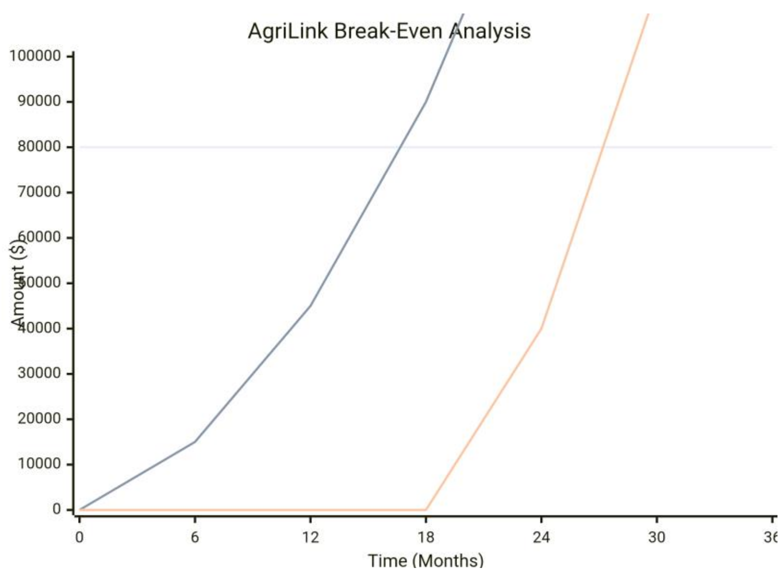
5.2.8.3 Other Costs

Item	Quantity	Cost (BDT)
Internet (Broadband)	6 months	6,000
Electricity	6 months	4,500
Domain Name (agrilink.com)	1 year	1,500
Hosting (Shared Server)	1 year	8,000
Printing & Documentation	–	2,000
Subtotal		**22,000**

5.2.9 Cost Benefit Analysis

Benefits (fair pricing, transparency) outweigh setup cost.





5.3 System Testing

5.3.1 Introduction

Testing validates correctness and security.

5.3.2 Test Plan

- Unit testing for each PHP file.
- Integration testing for workflows (signup → cart → checkout).
- User acceptance testing.

6. System Requirements & Using Guideline

6.1 System Requirements

6.1.1 Hardware Requirements

- Intel i5 processor or higher
- 8 GB RAM
- 250 GB HDD/SSD

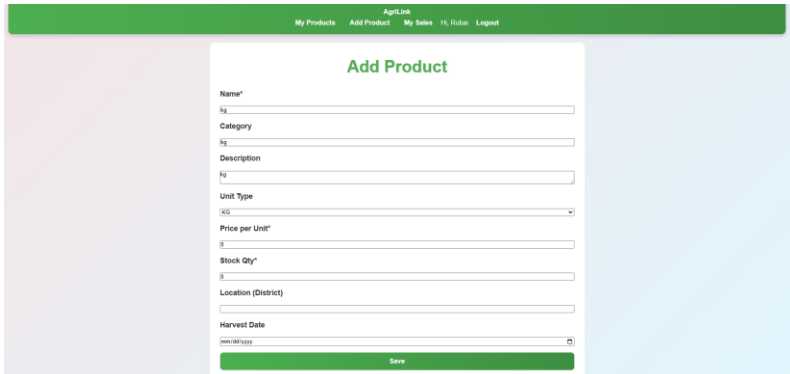
6.1.2 Software Requirements

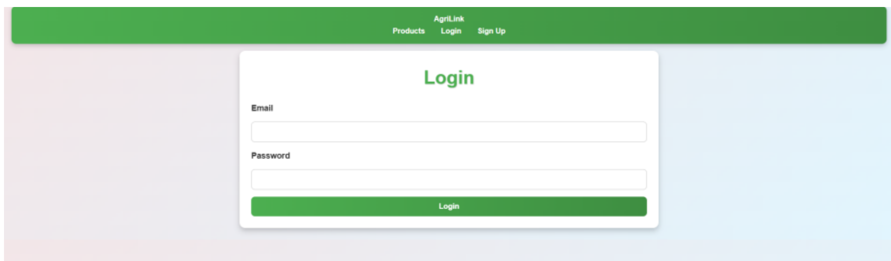
- XAMPP (PHP, MySQL, Apache)
- VS Code / Sublime Text
- Browser (Chrome/Firefox)

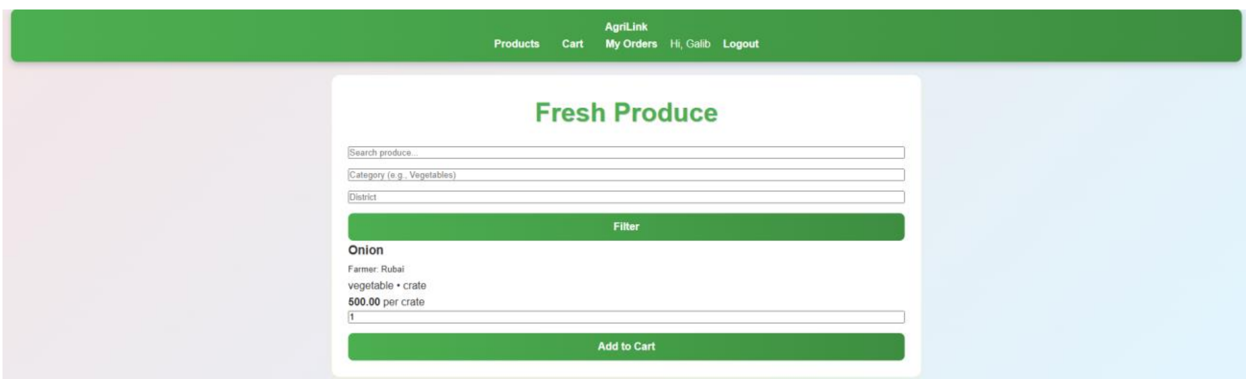
6.1.3 Using Guideline

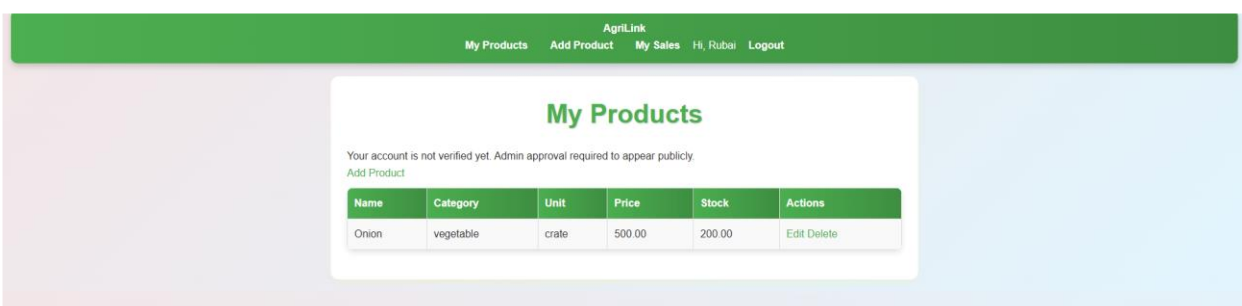
- Start Apache & MySQL from XAMPP.
- Place project folder inside `htdocs`.
- Import SQL file into phpMyAdmin.
- Access via `http://localhost/agrilink`.

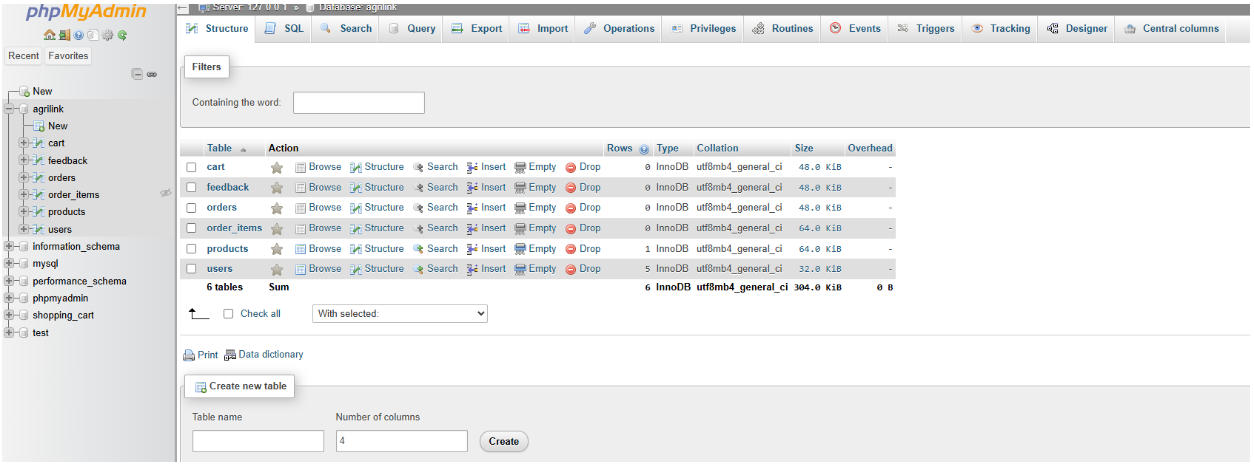
6.1.4 Screen Shots











7. Conclusion & Upcoming Features

7.1 Conclusion

The **AgriLink project** successfully demonstrates how a digital marketplace can eliminate intermediaries, giving farmers direct access to urban buyers. It integrates **authentication, database management, order processing, and admin controls**.

7.2 Upcoming Features

- Integration of online payment gateways (Bkash, Nagad, PayPal).
- AI-based product recommendations.
- Real-time delivery tracking.
- Mobile app version for farmers and buyers.

Information References

- PHP & MySQL Web Development (Luke Welling & Laura Thomson)
- Official PHP Documentation (https://www.php.net)
- MySQL Documentation (https://dev.mysql.com/doc/)
- TutorialsPoint PHP/MySQL

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